Three new problems from the Erdős corpus

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Abstract

We introduce three problems arising from a systematic study of the Erdős problem corpus. (1) *Density-relaxed Schur numbers* $\mathrm{DS}(k, \alpha)$ interpolate between Schur numbers and density Ramsey theory; we determine $\mathrm{DS}(2, \alpha)$ for all α, revealing a sharp phase transition at $\alpha = 1/2$, with a complete Lean 4 formalization. (2) *Coprime Ramsey numbers* $R_{\mathrm{cop}}(k)$ measure Ramsey forcing on the coprime graph; we prove $R_{\mathrm{cop}}(3) = 11$ by exhaustive extension. (3) We show that the set of primes, which maximizes $\sum 1/(a \log a)$ over primitive sets (Lichtman [5]), is *not* optimal for coprime pair counts, and give a lower bound $M^*(n) \geq \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}(1 + o(1))$.

MSC 2020: 05D10, 05C55, 11B75, 11N25.

1 Introduction

A systematic study of the 1,135 problems in the Erdős problem database [8] reveals natural questions at the intersection of Erdős's interests—Schur numbers, coprime graphs, primitive sets—that do not appear to have been formulated. We present three such problems with solutions and computational results. All code and Lean formalizations are publicly available.[1]

2 Density-relaxed Schur numbers

A *Schur triple* is a triple (a, b, c) with $a + b = c$ and $a, b \geq 1$ (so $a = b$ is permitted).

Definition 2.1. For $k \geq 1$ and $\alpha \in (0, 1]$, a k-coloring of $[N]$ is (k, α)-*avoiding* if every color class is both sum-free and has at most αN elements. Define

$$\mathrm{DS}(k, \alpha) \;=\; 1 + \max\{N \geq 1 : \text{a } (k, \alpha)\text{-avoiding coloring of } [N] \text{ exists}\},$$

with $\mathrm{DS}(k, \alpha) = 1$ if no such N exists.

When $\alpha = 1$ the density constraint is vacuous, so $\mathrm{DS}(k, 1) = S(k) + 1$ where $S(k)$ is the k-th Schur number [7]. When $\alpha < 1/k$, pigeonhole forces some class to exceed αN elements for all $N \geq 1$, so $\mathrm{DS}(k, \alpha) = 1$. Since Schur triples force at $S(k) + 1$ independently of density, we always have $\mathrm{DS}(k, \alpha) \leq S(k) + 1$.

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[1]Code and data: `doi.org/10.5281/zenodo.18638635`

Theorem 2.2. $\mathrm{DS}(2,\alpha) = 5$ *for all* $\alpha \geq 1/2$*, and* $\mathrm{DS}(2,\alpha) = 1$ *for* $\alpha < 1/2$*. In particular* $\mathrm{DS}(2,\frac{1}{2}) = 5 = S(2)+1$.

Proof. For $\alpha < 1/2$: in any 2-coloring of $[N]$, some class has $\lceil N/2 \rceil > \alpha N$ elements (since $\alpha < 1/2$), so no $(2,\alpha)$-avoiding coloring exists for any N.

For $\alpha \geq 1/2$: the coloring $C_1 = \{1,4\}$, $C_2 = \{2,3\}$ of [4] has both classes sum-free with $|C_i| = 2 \leq \alpha \cdot 4$ (using $\alpha \geq 1/2$). So $N = 4$ is $(2,\alpha)$-avoiding. Since $S(2) = 4$, every 2-coloring of [5] contains a Schur triple, so $N = 5$ is not avoiding. Hence $\mathrm{DS}(2,\alpha) = 1 + 4 = 5$. □

Remark 2.3. The density-forcing threshold ($\alpha = 1/2 = 1/k$ for $k = 2$) coincides with the Schur-forcing threshold ($S(2)+1 = 5$), collapsing $\mathrm{DS}(2,\cdot)$ to a step function. For $k \geq 3$, these thresholds may separate, yielding a non-trivial interpolation regime.

Remark 2.4. Theorem 2.2 has been fully formalized in Lean 4 (Mathlib), with 12 theorems proved and 0 `sorry` statements. The formalization is in `lean/NPG2_DensitySchur.lean`.

Open Problem 2.5. Determine $\mathrm{DS}(k,\alpha)$ for $k \geq 3$ and $\alpha \in (1/k, 1)$. Is there a non-trivial interpolation regime, i.e., does $\mathrm{DS}(k,\alpha) < S(k)+1$ for some $\alpha > 1/k$?

3 Coprime Ramsey numbers

Definition 3.1. The *coprime graph* $G_{\mathrm{cop}}(n)$ has vertex set $[n]$ with $\{i,j\}$ an edge iff $\gcd(i,j) = 1$. Define $R_{\mathrm{cop}}(k)$ as the minimum n such that every 2-coloring of $E(G_{\mathrm{cop}}(n))$ contains a monochromatic K_k.

Since $E(G_{\mathrm{cop}}(n)) \subseteq E(K_n)$, any 2-coloring of K_n restricts to a 2-coloring of $G_{\mathrm{cop}}(n)$; if the former avoids monochromatic K_k, so does the latter (any monochromatic K_k in $G_{\mathrm{cop}}(n)$ would use only coprime edges, hence appear in K_n too). Therefore $R_{\mathrm{cop}}(k) \geq R(k,k)$ [6].

The clique number satisfies $\omega(G_{\mathrm{cop}}(n)) = \pi(n)+1$: the set $\{1\} \cup \{\text{primes} \leq n\}$ is a clique of this size. For the upper bound, note that in any pairwise coprime set $S \subseteq \{2,\ldots,n\}$, each element's prime factorization uses a disjoint set of primes (else two elements share a factor); since each element uses ≥ 1 prime and all primes used are $\leq n$, we have $|S| \leq \pi(n)$.

Theorem 3.2. $R_{\mathrm{cop}}(3) = 11$.

Proof. By exhaustive extension from $n = 8$.

Base. Among all 2^{21} edge 2-colorings of $G_{\mathrm{cop}}(8)$ (21 edges), exactly 36 avoid monochromatic triangles.

Extension. For $n = 9$: vertex 9 adds 6 coprime edges; extending 36 colorings through 2^6 options each yields 36 avoiding. For $n = 10$: vertex 10 adds 4 edges; extension yields 156 avoiding. For $n = 11$: vertex 11 is prime, adding 10 edges (coprime to all of $1,\ldots,10$); extending 156 colorings through 2^{10} options each yields 0 avoiding.

n	Edges in $G_{\mathrm{cop}}(n)$	New edges	Avoiding colorings
8	21	—	36
9	27	6	36
10	31	4	156
11	41	10	0

□

The increase at $n = 10$ reflects that $10 = 2 \times 5$ has only 4 coprime neighbors, imposing weak constraints. The collapse at $n = 11$ is driven by 11 being prime: its 10 edges create too many triangles for any 2-coloring to avoid.

Remark 3.3. $R_{\text{cop}}(3)/R(3,3) = 11/6 \approx 1.83$ exceeds the inverse coprime density $\pi^2/6 \approx 1.64$. If $G_{\text{cop}}(n)$ behaved like a random graph of density $6/\pi^2$, one would predict $R_{\text{cop}}(3) \approx 10$. The excess arises from multiplicative correlations among missing edges (multiples of a prime p form correlated non-adjacencies) [4].

Conjecture 3.4. $R_{\text{cop}}(4) = 20$. Evidence: *An avoiding coloring at $n = 19$ was found by random search ($\sim$160,000 trials); no avoiding coloring was found at any $n \in \{20, \ldots, 49\}$ ($\sim 10^6$ random trials each). This is heuristic evidence only—the extension method is infeasible at this scale ($\sim 1.47 \times 10^6$ base colorings).*

Open Problem 3.5. Determine the growth rate of $R_{\text{cop}}(k)$. Is $R_{\text{cop}}(k) > R(k,k)$ for all $k \geq 3$?

4 Coprime pairs in primitive sets

A set $A \subseteq [n]$ is *primitive* if no element divides another [2]. By Dilworth's theorem [1], every primitive $A \subseteq [n]$ has $|A| \leq \lceil n/2 \rceil$. Let $M(A) = |\{(a,b) \in A^2 : a < b,\ \gcd(a,b) = 1\}|$ and $M^*(n) = \max\{M(A) : A \subseteq [n] \text{ primitive}\}$.

Lichtman [5] proved that primes maximize $f(A) = \sum_{a \in A} 1/(a \log a)$ over primitive A—the Erdős primitive set conjecture. We show primes do *not* maximize M.

The *top layer* $T(n) = \{\lfloor n/2 \rfloor + 1, \ldots, n\}$ is primitive with $|T(n)| = \lceil n/2 \rceil$ (Dilworth-maximum). Define the *shifted top layer* $S(n)$ by replacing each even $2k \in T(n)$ with k when k is odd and $k > n/3$.

Lemma 4.1. *$S(n)$ is primitive with $|S(n)| = \lceil n/2 \rceil$.*

Proof. Replacement is bijective (each $2k$ maps to a distinct $k \in (n/3, n/2] \setminus T(n)$), preserving size. For primitivity: retained elements of $T(n)$ exceed $n/2$, so no two divide each other. A new element $k \in (n/3, n/2]$ cannot divide a retained element $b > n/2$: the multiple $2k$ was removed, and $3k > n \geq b$. Two new elements $k_1 < k_2$ with $k_1 \mid k_2$ would require $k_2 \geq 2k_1 > 2n/3 > n/2 \geq k_2$, a contradiction. □

Proposition 4.2. *As $n \to \infty$, $M(T(n)) \sim \frac{6}{\pi^2}\binom{\lceil n/2 \rceil}{2}$ and $M(S(n)) \sim \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}$.*

Proof. For $T(n)$: the proportion of coprime pairs among L consecutive integers converges to $\prod_p (1 - 1/p^2) = 6/\pi^2$ as $L \to \infty$ (classical; see, e.g., Hardy–Wright [3], Thm. 332). Since $T(n)$ consists of $\lceil n/2 \rceil$ consecutive integers, the claim follows.

For $S(n)$: the even fraction satisfies $f_E \to 1/3$ (we replace roughly 1/6 of the $\lceil n/2 \rceil$ elements, reducing even elements from 1/2 to 1/3). Two even elements share factor 2, contributing 0 coprime pairs. For pairs with at least one odd element, coprimality is governed by odd primes; since the elements of $S(n)$ are drawn from intervals of length $\Theta(n)$ in $[n]$, the coprime probability for such pairs converges to $\prod_{p \geq 3}(1 - 1/p^2) = 8/\pi^2$. Weighting by parity classes: $d(S(n)) \to \frac{8}{\pi^2}(1 - f_E^2) = \frac{64}{9\pi^2}$. □

Theorem 4.3. *$M^*(n) \geq \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}(1 + o(1))$. For $n \geq 10$, $M(P(n)) < M^*(n)$ where $P(n)$ is the set of primes up to n; the latter claim is verified by exhaustive computation.*

Proof. The lower bound is realized by $S(n)$ (Lemma 4.1 and Proposition 4.2). Since $M(P(n)) = \binom{\pi(n)}{2} \sim n^2/(2\ln^2 n) = o(n^2)$ while $M(S(n)) = \Theta(n^2)$, primes are eventually suboptimal. Exhaustive computation confirms this for $n \geq 10$:

n	Optimal primitive set	$M^*(n)$	$M(P(n))$
8	$\{2, 3, 5, 7\}$	6	6
10	$\{4, 5, 6, 7, 9\}$	8	6
15	$\{4, 6, 7, 9, 10, 11, 13, 15\}$	21	15
18	$\{4, 6, 7, 9, 10, 11, 13, 15, 17\}$	29	21

□

Conjecture 4.4. $M^*(n) \sim \frac{64}{9\pi^2}\binom{\lceil n/2 \rceil}{2}$. *That is, $S(n)$ is asymptotically optimal among primitive sets for maximizing coprime pairs.*

Remark 4.5. Optimal sets use prime powers ($4 = 2^2$, $9 = 3^2$) and products ($6 = 2 \cdot 3$, $10 = 2 \cdot 5$) instead of small primes. Primes have coprime density $d = 1$ but size only $\pi(n)$; Dilworth-maximum sets have size $\lceil n/2 \rceil$ but lower density. Since $M \propto d \cdot |A|^2$, size dominates.

Open Problem 4.6. Characterize which functionals F on primitive sets are maximized by primes. Lichtman's $\sum 1/(a \log a)$ favors primes; $M(A)$ does not.

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